

Phishing Simulation using Gophish

This document demonstrates the setup and execution of a phishing simulation using Gophish in a controlled educational environment for cybersecurity learning purposes.

Abstract

This project demonstrates how a phishing awareness simulation can be created using Gophish on Kali Linux. The objective of the project is to understand how phishing campaigns work, how attackers attempt credential harvesting, and how organizations can use simulations to train users against phishing attacks. The project includes the setup of Gophish, configuration of SMTP settings, creation of landing pages, email templates, launching campaigns, and monitoring captured results.

Introduction

Phishing is one of the most common forms of social engineering attacks where attackers attempt to trick users into revealing sensitive information such as usernames and passwords. Gophish is an open-source phishing framework used for security awareness training and simulations. This project focuses on understanding phishing mechanisms ethically in a controlled lab environment.

Objectives

- Understand phishing and social engineering techniques.
- Learn how Gophish works.
- Configure phishing campaigns in a safe environment.
- Analyze campaign statistics and user interactions.
- Gain practical cybersecurity experience.

Tools and Technologies Used

- Kali Linux
- Gophish
- Gmail SMTP
- Virtual Machine
- Web Browser

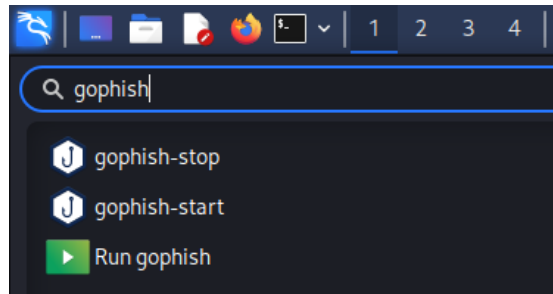
System Requirements

- Minimum 4 GB RAM
- VirtualBox or VMware

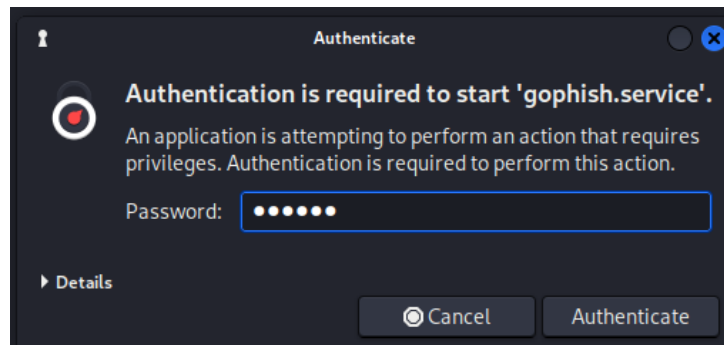
- Kali Linux installed
- Internet connection
- Gmail account with App Password enabled

Step 1: Launching Gophish

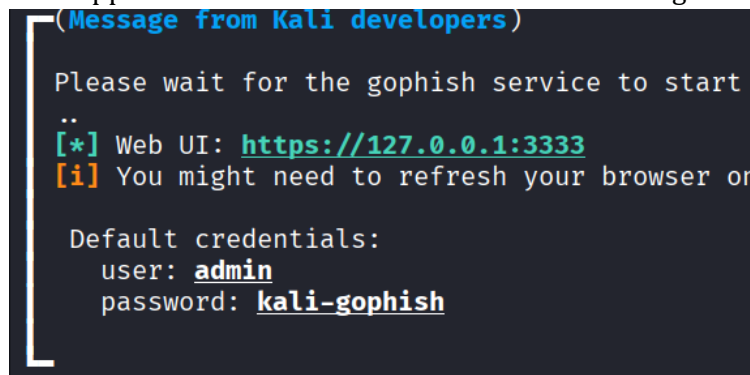
- A. Start Kali Linux Virtual Machine on your system. Search for gophish and click on gophish-start



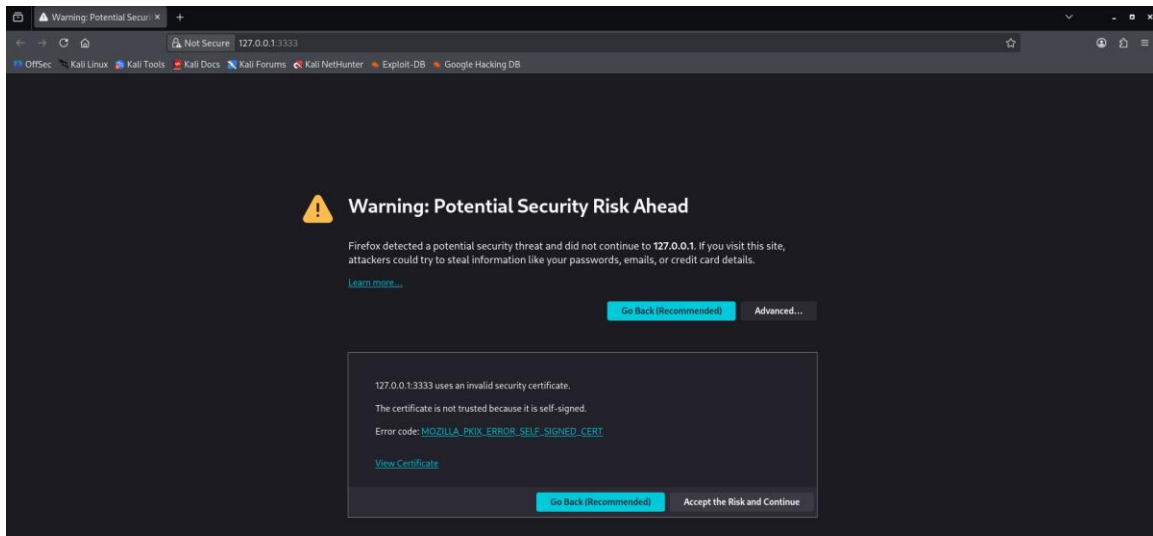
- B. Enter the password to start the service



- C. Click the link that appears in the terminal and use the credentials given to log in.



- D. A warning is received on the browser. Click on Advanced>Accept risk and continue.



- E. After this, you will be redirected to the Gophish login page. Enter the credentials given in the terminal to log in.



Please sign in

Sign in

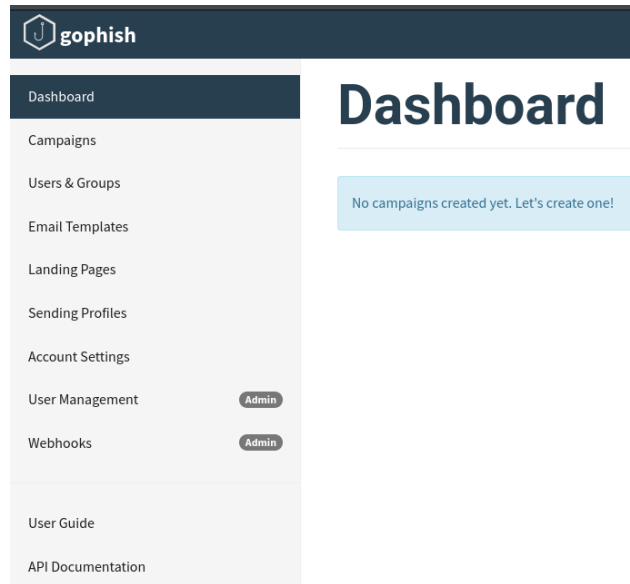
- F. Then reset your password



Reset Your Password

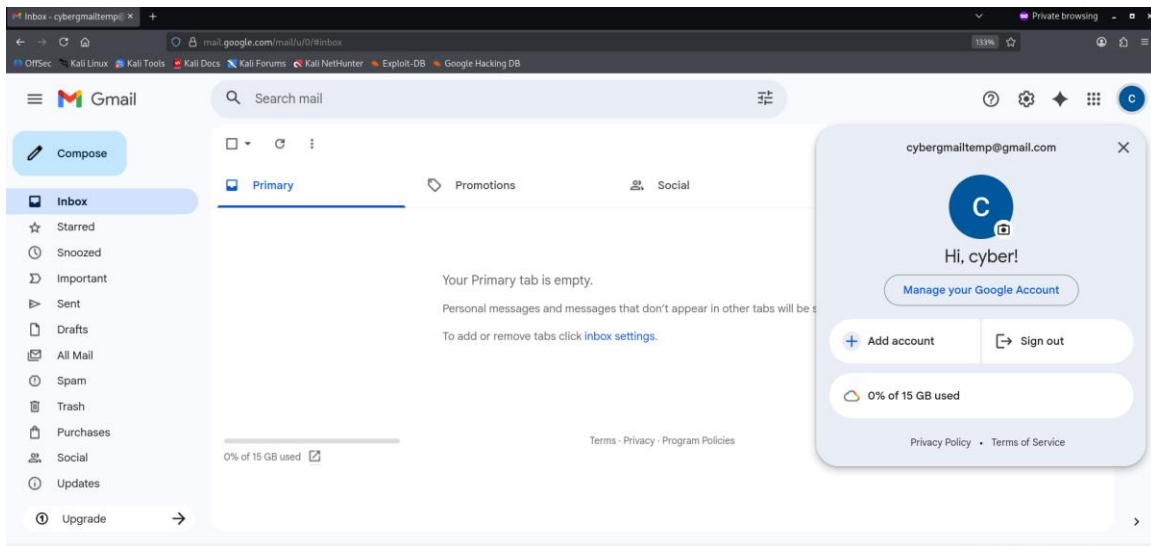
Save Password

G. You are now logged in to the Gophish webpage.

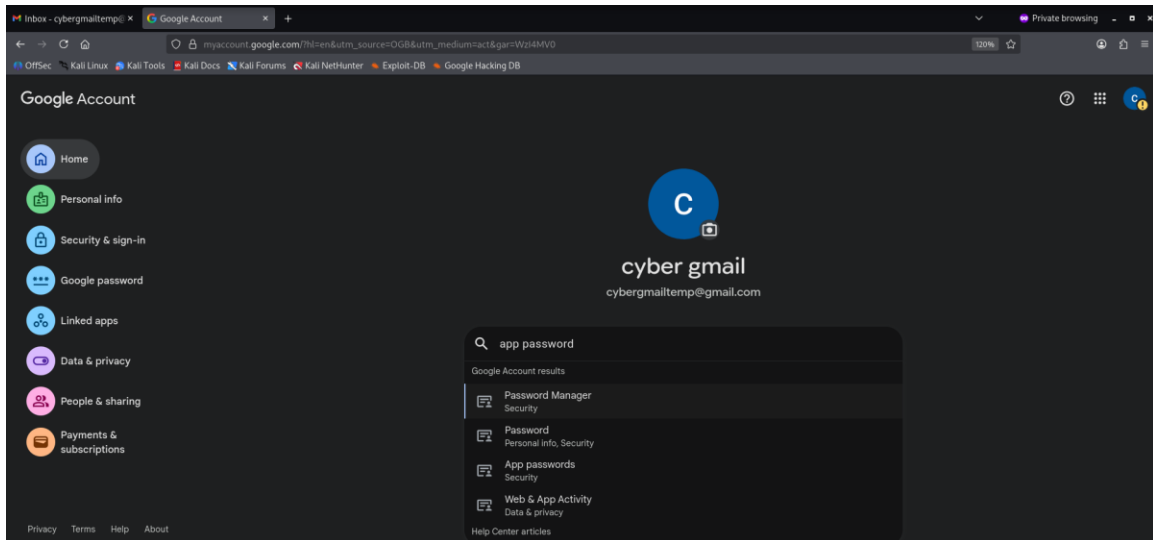


Step 2: Setting up Gmail account

A. Log into your gmail account. Click on Manage Your Google Account.



B. Search for App Password and click on it. It will only appear if you have 2-step verification turned on.



C. Create an app password and name it GoPhish.

← App passwords

App passwords help you sign into your Google Account on older apps and services that don't support modern security standards.

App passwords are less secure than using up-to-date apps and services that use modern security standards. Before you create an app password, you should check to see if your app needs this in order to sign in.

[Learn more](#)

You don't have any app passwords.

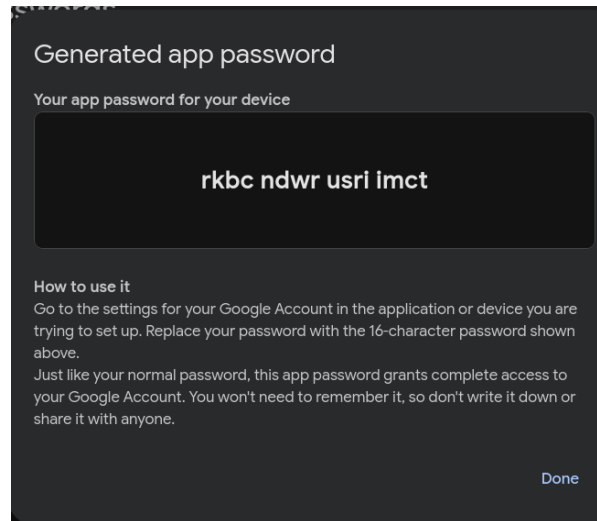
To create a new app specific password, type a name for it below...

App name

GoPhish

Create

D. Copy the app password you get.



Step 3: Configuring Sending Profile

- A. In Gophish webpage, go to Sending Profile. Click on New to create a Sending profile with which emails will be sent. You will get a popup like the one below. Enter all details (adjust as needed). Here, we are using a Gmail SMTP server. In the password field, enter the password you copied in the last step.

A screenshot of the "New Sending Profile" form in Gophish. The form includes fields for Name, Interface Type (SMTP), SMTP From, Host, Username, Password, and a checkbox for "Ignore Certificate Errors". There is also a section for "Email Headers" with a table and a "Send Test Email" button at the bottom.

B. Click on Send Test Email. Enter details as shown below and click on Send.

Send Test Email

✔ Email Sent!

Send Test Email to:

cyber

gmail

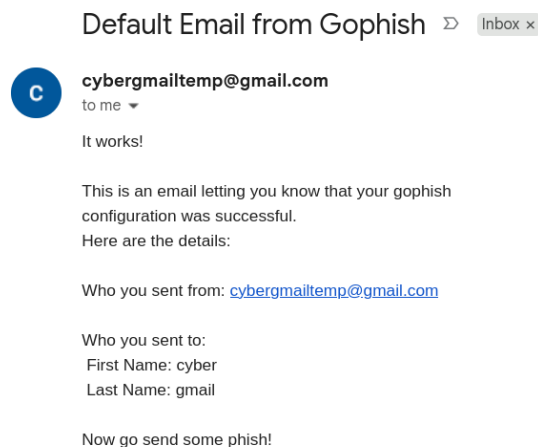
cybergmailtemp@gmail.com

Position

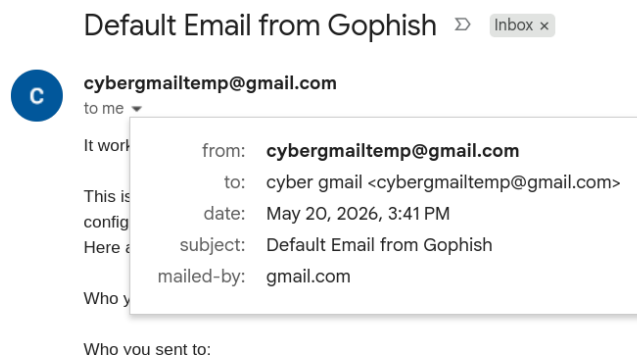
Cancel

Send

C. Go back to your gmail account to check if the mail was received. If not received, check previous steps for any mistakes.

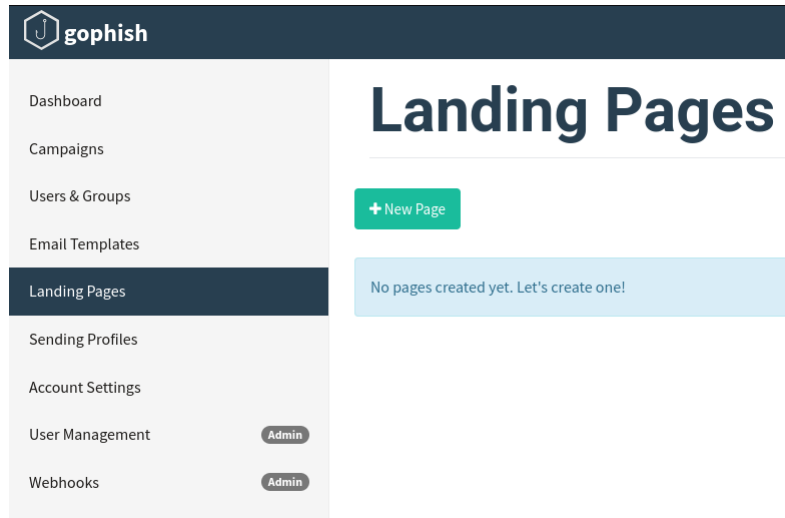


D. You can verify mail authenticity by clicking on the small arrow below sender gmail. Your sending profile has been set up.



Step 3: Creating Landing Page

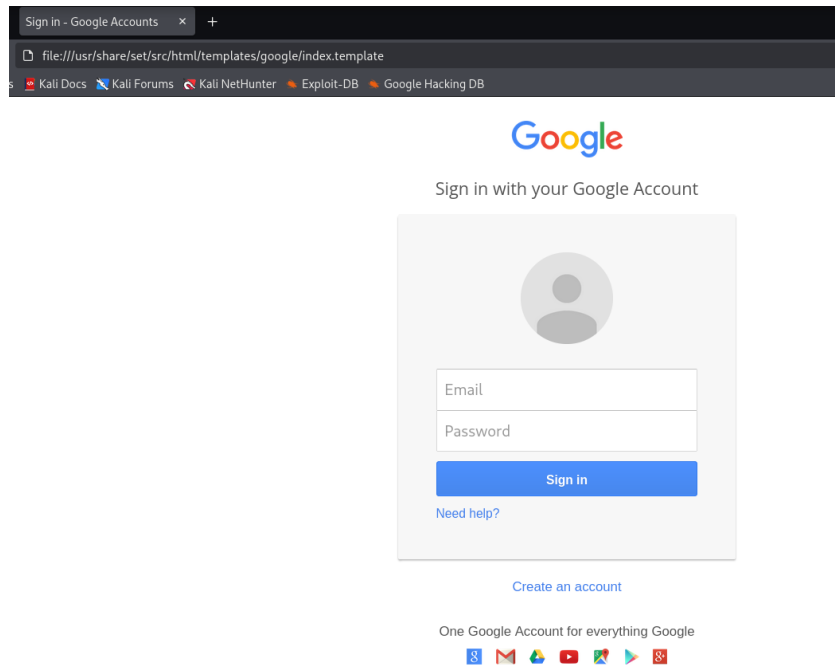
A. In Gophish webpage, click on Landing Pages.



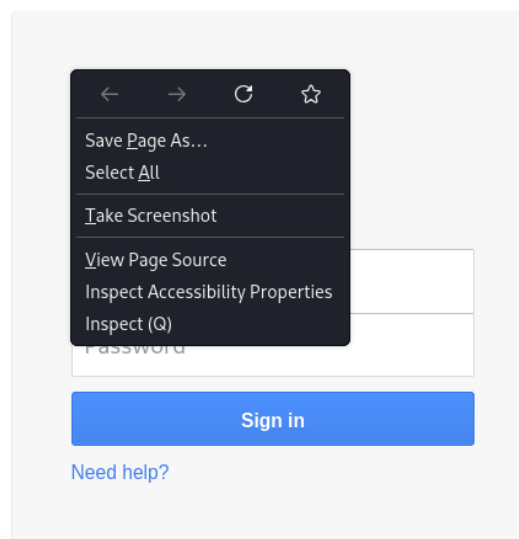
B. In terminal, perform the commands as shown below. We use SET to get the google login page template to use as your phishing login page to make it look authentic.

```
(kali㉿kali)-[~]  
$ cd /usr/share/set/src/html/templates/google  
  
(kali㉿kali)-[/usr/.../src/html/templates/google]  
$ ls  
index.template  
  
(kali㉿kali)-[/usr/.../src/html/templates/google]  
$ open index.template  
  
(kali㉿kali)-[/usr/.../src/html/templates/google]  
$
```

C. This opens a webpage as shown below.



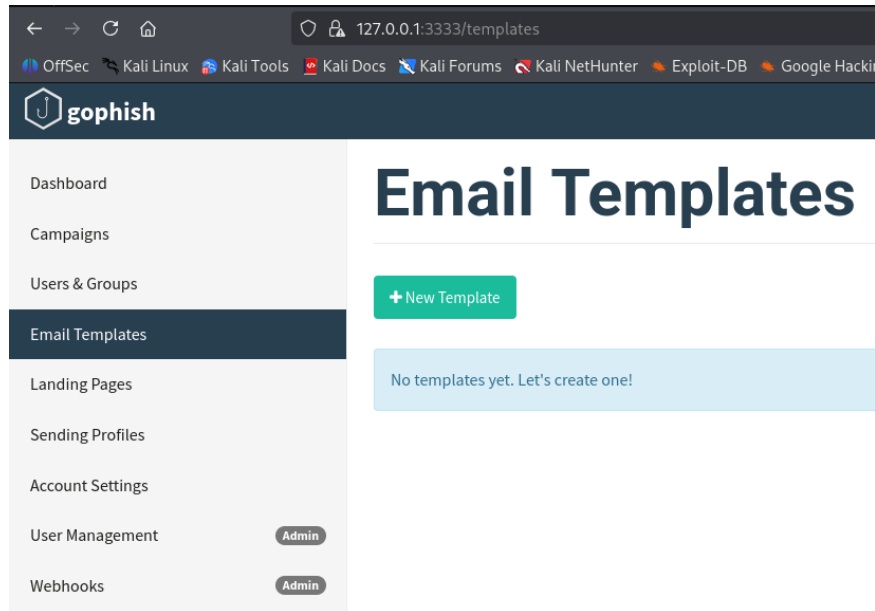
D. Right click on the login page and then click on View Page Source.



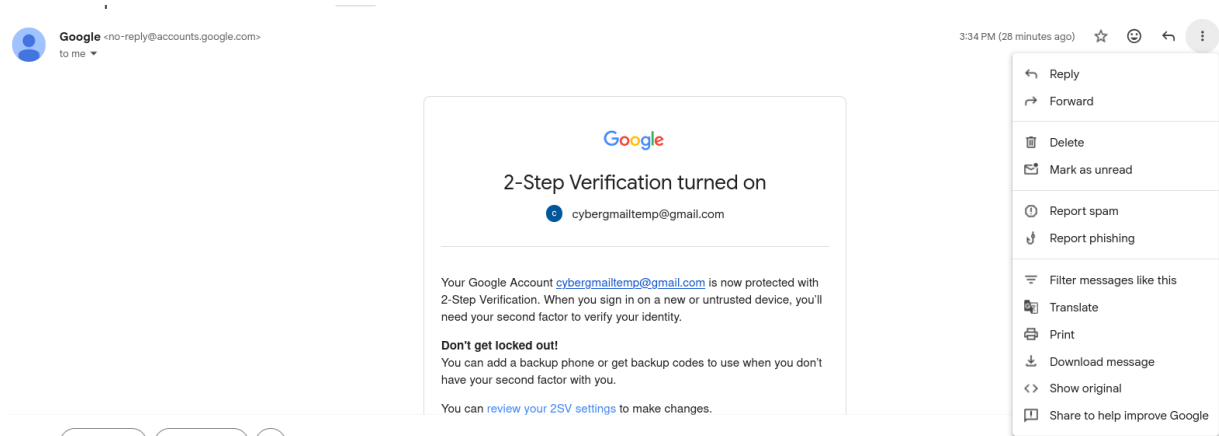
E. You will be redirected to HTML code. Select the whole code by using Ctrl+A and then copy.

Step 4: Creating Email Template

A. In Gophish, click on Email Templates.



B. In Gmail, log into your google account and open a google mail which looks authentic. The goal is to make the user think it is an actual email from google. Here, an email for 2-step verification activation is used. Click on the three dots on the top right and then click on Show Original.



C. Click Copy To Clipboard.

Original Message

Message ID	<jcx9FIV7DII9kDahgOVvBQ@notifications.google.com>
Created at:	Wed, May 20, 2026 at 3:34 PM (Delivered after 1 second)
From:	Google <no-reply@accounts.google.com>
To:	cybergmailtemp@gmail.com
Subject:	2-Step Verification turned on
SPF:	PASS with IP 209.85.220.73 Learn more
DKIM:	'PASS' with domain accounts.google.com Learn more
DMARC:	'PASS' Learn more

[Download Original](#)

[Copy to clipboard](#)

D. In Gophish webpage, click on New Email Template. Click on Import Email.

New Template

Name:

Envelope Sender:

Subject:

☒ Text ☐ HTML

Plaintext

☒ Add Tracking Image

E. A popup will appear. Paste the whole code you copied earlier. Click on Import. Make sure that the checkbox is ticked.

Import Email

Email Content:

```

ba(0,0,0,0.54); font-size: 11px; line-height: 18px; padding-top: 12px; text-align: center;">
<div>You received this email to let you know about important changes to your Google Account and services.</div>
<div style=3D"direction: ltr;">
&copy; 2026 Google LLC,
<a class=3D"afal" style=3D"font-family: Roboto-Regular, Helvetica, Arial, sans-serif; color: rgba(0,0,0,0.54); font-size: 11px; line-height: 18px; padding-top: 12px; text-align: center;">
1600 Amphitheatre Parkway, Mountain View, CA 94043, USA</a>
</div>
</div>
</td>
<td width=3D"8" style=3D"width: 8px;">
</td>
</tr>
</table>
</td>
</tr>
<tr height=3D"32" style=3D"height: 32px;">
<td>
</td>
</tr>
</table>
</body>
</html>
--000000000000bfe889065244e25b--

```

☒ Change Links to Point to Landing Page

Cancel

Import

F. Fill the rest of the details as shown below. Adjust as needed.

New Template

Name:

Google

✉ Import Email

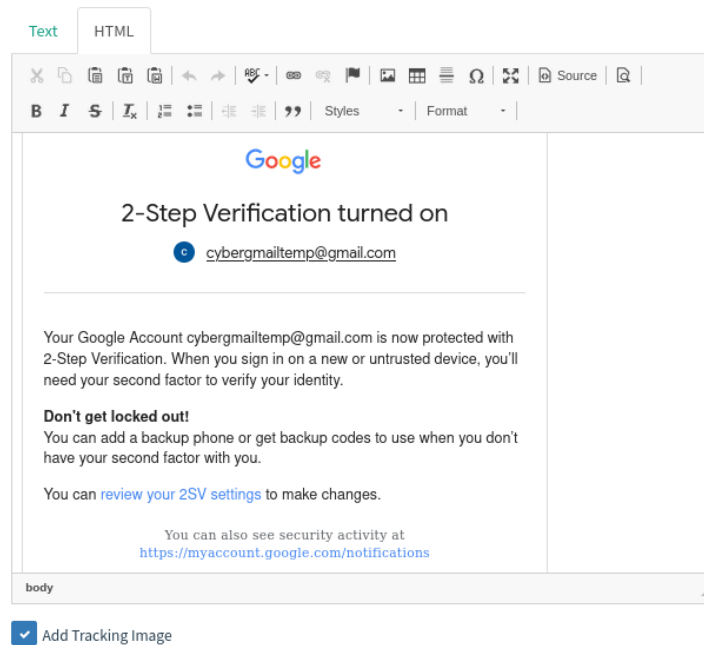
Envelope Sender: ⓘ

cybergmailtemp123@gmail.com

Subject:

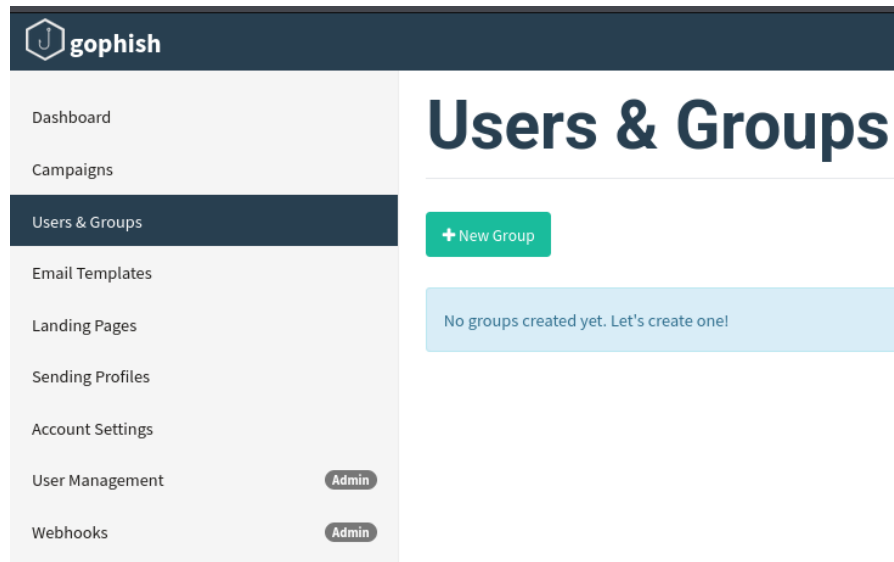
CAUTION: 2-STEP VERIFICATION TURNED ON!!!

G. You can see the preview of your phishing email template in the same window popup as shown below.



Step 5: Adding Target Users

- A. To now create a list of accounts to send phishing mails to, go to Users And Groups in the Gophish webpage.



- B. Fill details as shown below. You can add accounts manually or import a list. These will be the targets who will receive the phishing mails.

New Group

Name:

Test 1

+ Bulk Import Users

Download CSV Template

First Name

Last Name

Email

Position

+ Add

Show 10 entries

Search:

First Name	Last Name	Email	Position
cyber	gmail	cybergmailtem...	

Showing 1 to 1 of 1 entries

Previous

1

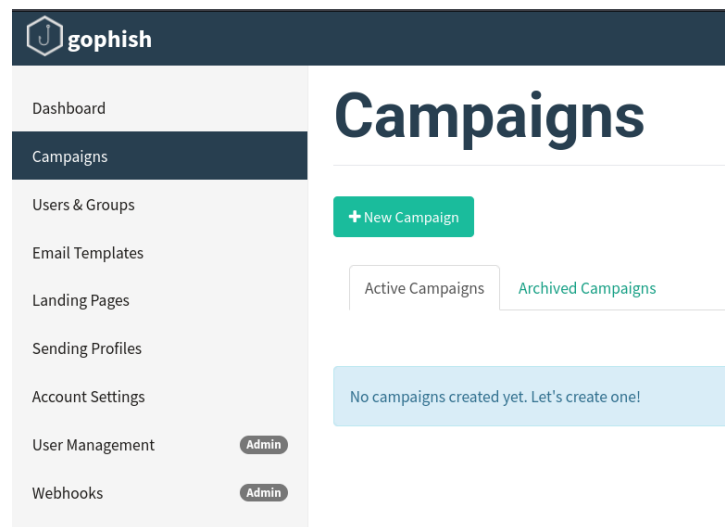
Next

Close

Save changes

Step 6: Launching Campaign

- A. To start the phishing process, go to Campaigns in Gophish webpage.



- B. Fill information as shown below and adjust as needed. In URL, enter the IP address of the virtual machine hosting GoPhish. Click on Launch Campaign.

Name:

Email Template:

Landing Page:

URL: 

Launch Date Send Emails By (Optional) 

Sending Profile:
 

Groups:

C. Click Launch again.



Are you sure?

This will schedule the campaign to be launched.

D. Campaign has launched successfully.



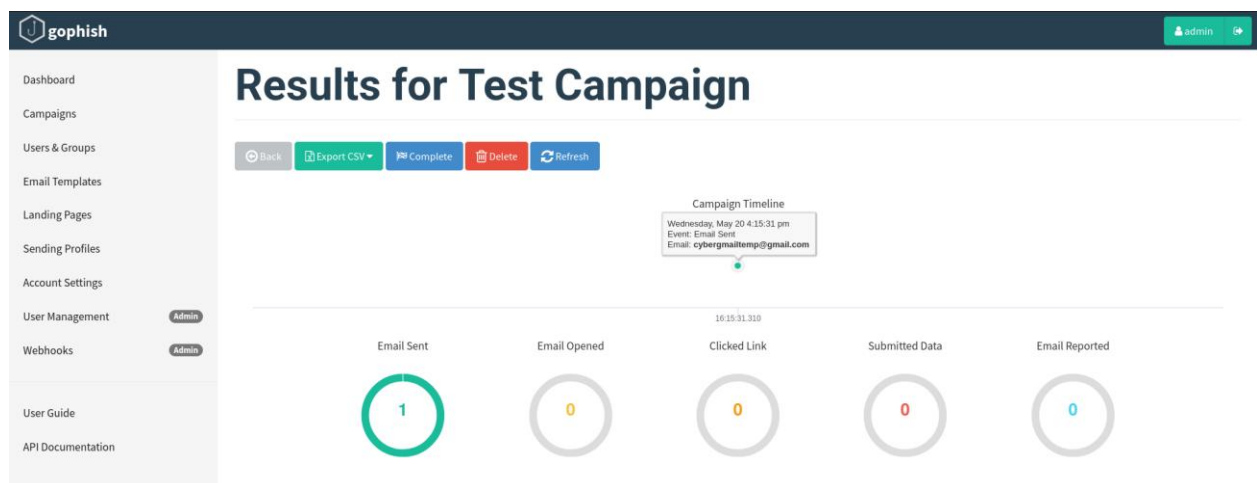
Campaign Scheduled!

This campaign has been scheduled for launch!

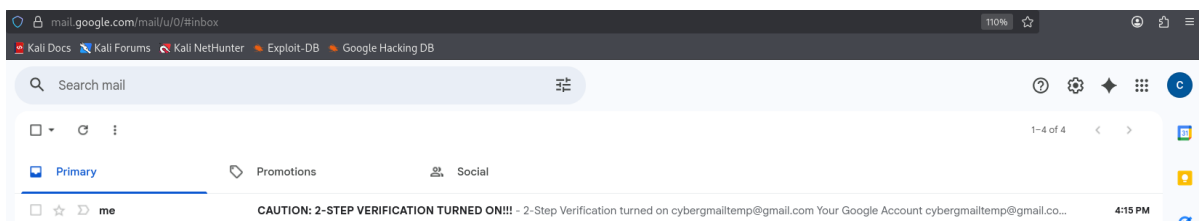
OK

Step 7: Monitoring Results

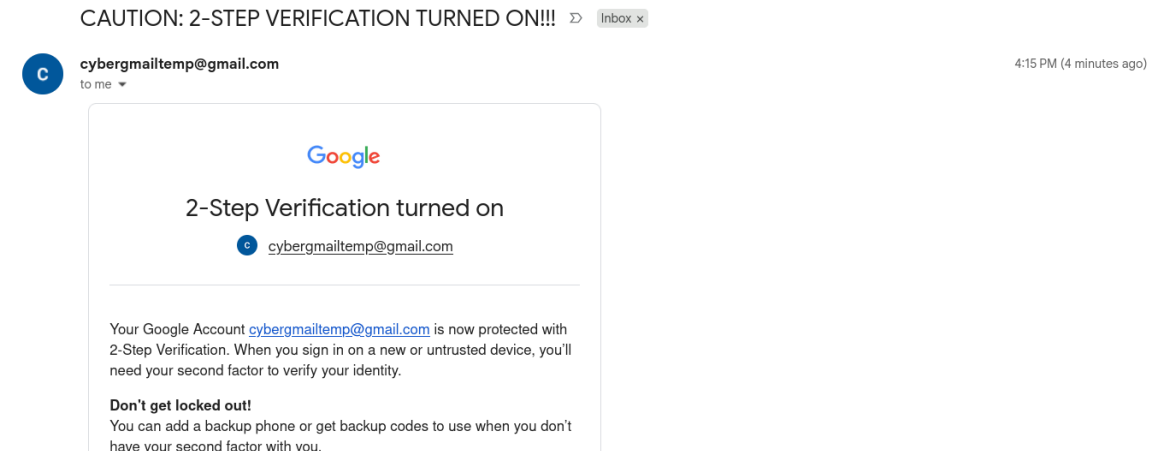
- A. After clicking on OK, you will get redirected to the campaign dashboard where you can track the phishing process. In the image, it shows that an email has been sent. This can be verified by checking the recipient's gmail account.



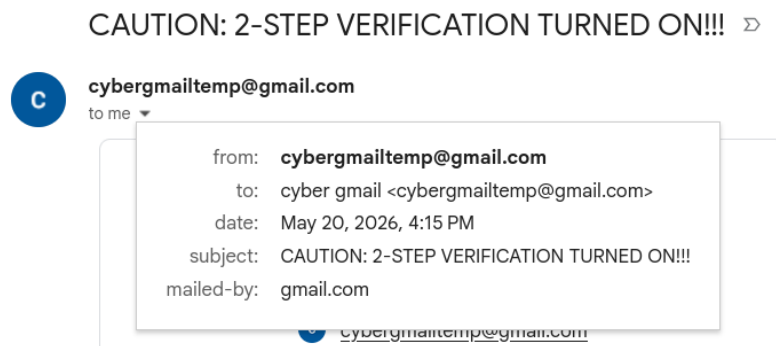
- B. After logging into the gmail account, we can see that a mail has been received.



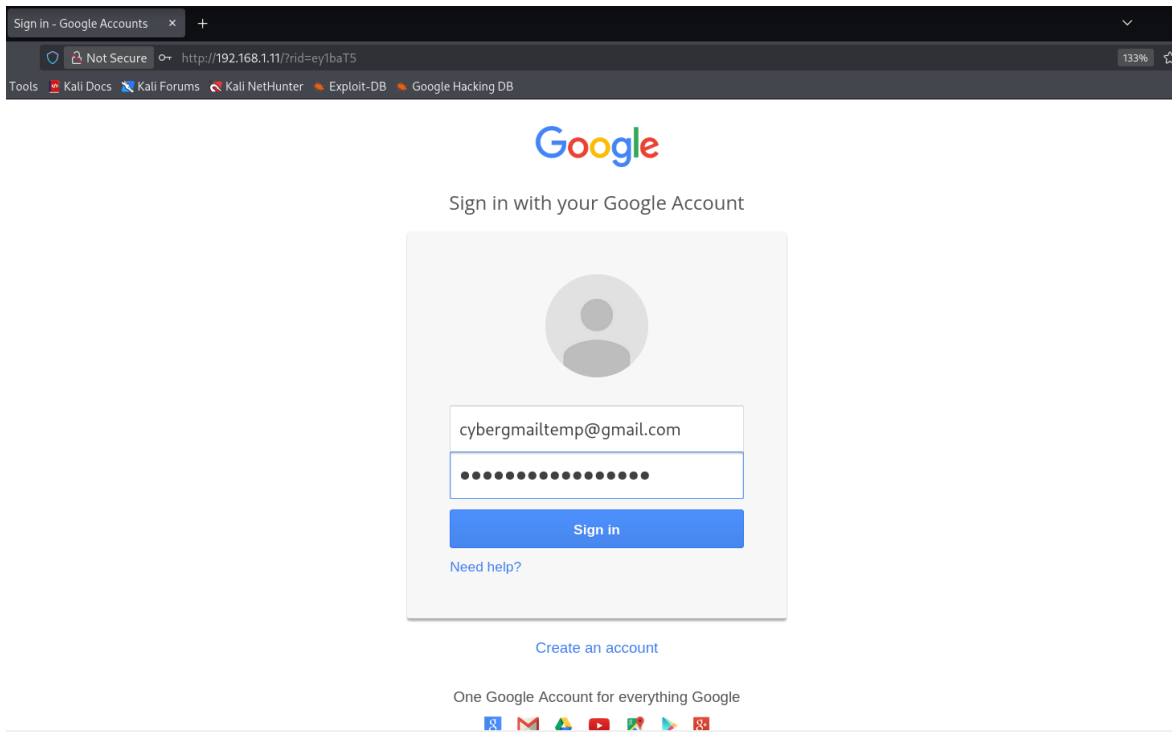
C. Click on the mail and click on the link present in the underlined gmail account.



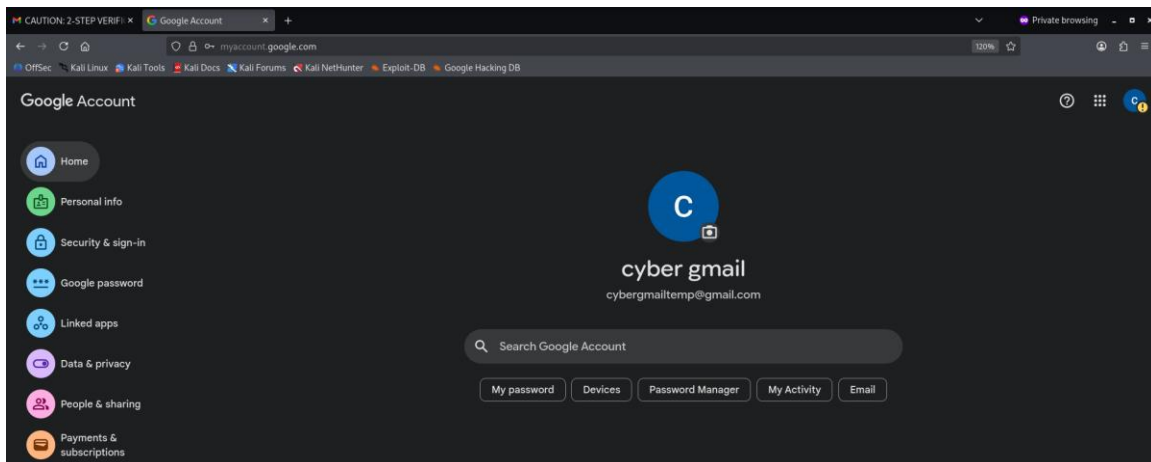
D. Authenticity of gmail can be verified by clicking the small arrow below the sender gmail address.



E. After clicking the phishing link, we get redirected to the login page. You can recognise that it is a phishing link by the “Not Secure” sign next to the link in the search tab. Enter your username and password to complete the phishing process.

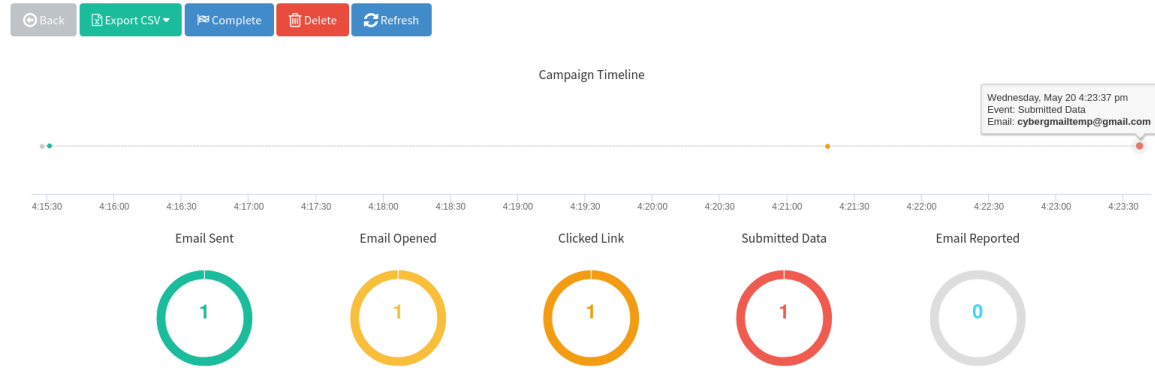


E. After clicking on Sign In, you will be redirected to your account page.



F. In Gophish webpage, we can see the updated phishing process. Email has been sent, opened, link clicked and details submitted.

Results for Test Campaign



G. Scroll down and click on the small arrow next to the target to see further details.

First Name	Last Name	Email	Position	Status	Reported
cyber	gmail	cybergmailtemp@gmail.com		Submitted Data	

Timeline for cyber gmail

Email: cybergmailtemp@gmail.com
Result ID: ey1ba75

- Campaign Created May 20th 2026 4:15:27 pm
- Email Sent May 20th 2026 4:15:31 pm
- Clicked Link May 20th 2026 4:21:18 pm
 - Linux (OS Version: x86_64)
 - Firefox (Version: 140.0)
- Submitted Data May 20th 2026 4:23:37 pm
 - Linux (OS Version: x86_64)
 - Firefox (Version: 140.0)

Replay Credentials

View Details

H. Click on View Details at the end to see login username and password. Phishing process has been successfully completed. Click on Complete Campaign.

▼ View Details

Parameter	Value(s)
Email	cybergmailtemp@gmail.com
GALX	SJLCKfgaqoM
PersistentCookie	yes
_utf8	Ⓜ
bgresponse	js_disabled
checkConnection	
checkedDomains	youtube
continue	https://accounts.google.com/o/oauth2/auth?zt=ChRsWFBwd2JmV1hlcDhtUFdlldzBENhlfVWsxSTdNLW9MdThibW1TMFQzVUZFc1BBaURuWmlRSQ%E2%88%99APsBz4gAAAAUy4_qD7Hbfz38w8kxnaNouLcRiD3YTjX
dnConn	
dsh	-7381887106725792428
password	cybergmailtemp123
pstMsg	1
service	Iso
signIn	Sign in

Results

The phishing simulation successfully demonstrated how phishing attacks are conducted and monitored. The project showed how user interactions can be tracked and analyzed for security awareness training.

Advantages

- Helps understand phishing techniques.
- Useful for cybersecurity awareness training.
- Easy to configure and monitor.
- Open-source and lightweight.

Limitations

- Requires ethical and legal authorization.
- SMTP providers may block suspicious activity.
- Not suitable for real-world unauthorized testing.

Ethical Considerations

This project was performed only for educational and awareness purposes in a controlled environment. No unauthorized access or malicious activity was performed. The project follows ethical cybersecurity practices.

Conclusion

The project provided hands-on experience with phishing simulations using Gophish. It improved understanding of phishing attacks, email spoofing concepts, landing page creation, and campaign monitoring. The project also highlighted the importance of cybersecurity awareness and ethical hacking practices.